



Table of Contents

1. [Cover](#)
2. [Table of Contents](#)
3. [Title Page](#)
4. [Copyright](#)
5. [Dedication](#)
6. [Biographies](#)
7. [Preface: QuantConnect](#)
8. [Introduction](#)
9. [Part I Foundations of Capital Markets and Quantitative Trading](#)
 1. [Chapter 1 Foundations of Capital Markets](#)
 1. [Market Mechanics](#)
 2. [Market Participants](#)
 3. [Data and Data Feeds](#)
 4. [Brokerages and Transaction Costs](#)
 5. [Security Identifiers](#)
 6. [Assets and Derivatives](#)
 2. [Chapter 2 Foundations of Quantitative Trading](#)
 1. [Research Process](#)
 2. [Testing and Debugging Tools](#)
 3. [Time and Look-ahead Bias](#)
 4. [Strategy Styles](#)
 5. [Parameter Sensitivity Testing and Optimization](#)
 6. [Margin Modeling](#)
 7. [Diversification and Asset Selection](#)
 8. [Indicators and Other Data Transformations](#)
 9. [Sourcing Ideas](#)
10. [Part II Foundations of AI and ML in Algorithmic Trading](#)

1. [Chapter 3 Step 1: Problem Definition](#)
2. [Chapter 4 Step 2: Dataset Preparation](#)
 1. [Data Collection](#)
 2. [Exploratory Data Analysis](#)
 3. [Data Preprocessing](#)
 4. [Feature Selection](#)
 5. [Splitting of Dataset into Training, Testing, and Possibly Validation Sets](#)
3. [Chapter 5 Step 3: Model Choice, Training, and Application](#)
 1. [Regression](#)
 2. [Classification](#)
 3. [Ranking](#)
 4. [Clustering](#)
 5. [Language Models](#)
11. [Part III Advanced Applications of AI in Trading and Risk Management](#)
 1. [Chapter 6 Applied Machine Learning](#)
 1. [Example 1—ML Trend Scanning with MLFinlab](#)
 2. [Example 2—Factor Preprocessing Techniques for Regime Detection](#)
 3. [Example 3—Reversion vs. Trending: Strategy Selection by Classification](#)
 4. [Example 4—Alpha by Hidden Markov Models](#)
 5. [Example 5—FX SVM Wavelet Forecasting](#)
 6. [Example 6—Dividend Harvesting Selection of High-Yield Assets](#)
 7. [Example 7—Effect of Positive-Negative Splits](#)
 8. [Example 8—Stop Loss Based on Historical Volatility and Drawdown Recovery](#)
 9. [Example 9—ML Trading Pairs Selection](#)
 10. [Example 10—Stock Selection through Clustering Fundamental Data](#)
 11. [Example 11—Inverse Volatility Rank and Allocate to Future Contracts](#)
 12. [Example 12—Trading Costs Optimization](#)

13. [Example 13—PCA Statistical Arbitrage Mean Reversion](#)
14. [Example 14—Temporal CNN Prediction](#)
15. [Example 15—Gaussian Classifier for Direction Prediction](#)
16. [Example 16—LLM Summarization of Tiingo News Articles](#)
17. [Example 17—Head Shoulders Pattern Matching with CNN](#)
18. [Example 18—Amazon Chronos Model](#)
19. [Example 19—FinBERT Model](#)
2. [Chapter 7 Better Hedging with Reinforcement Learning](#)
 1. [Introduction](#)
 2. [Overview of the Reinforcement Learning](#)
 3. [Implementation on QuantConnect](#)
 4. [Results](#)
 5. [Conclusion](#)
3. [Chapter 8 AI for Risk Management and Optimization](#)
 1. [What Is Corrective AI and Conditional Parameter Optimization?](#)
 2. [Feature Engineering](#)
 3. [Applying Corrective AI to Daily Seasonal Forex Trading](#)
 4. [What Is Conditional Parameter Optimization?](#)
 5. [Applying Conditional Parameter Optimization to an ETF Strategy](#)
 6. [Unconditional vs. Conditional Parameter Optimizations](#)
 7. [Performance Comparisons](#)
 8. [Conditional Portfolio Optimization](#)
 9. [Conclusion](#)
 10. [Definitions of Spread EMA & Spread VAR](#)
4. [Chapter 9 Application of Large Language Models and Generative AI in Trading](#)
 1. [Role of Generative AI in Creating Alpha](#)
 2. [Selecting an LLM for Building a Generative AI Application](#)
 3. [Prompt Engineering](#)
 4. [Question Answering Using a Retrieval Augmented Application in SageMaker Canvas](#)
 5. [Summarization](#)
 6. [Useful AI Platforms and Services](#)

12. [References](#)
13. [Subject Index](#)
14. [Code Index](#)
15. [End User License Agreement](#)

List of Illustrations

1. Chapter 1

1. [Figure 1.1 Flow of retail and institutional traffic across public and private markets, an...](#)
2. [Figure 1.2 Relative performance of various assets during the market crash on May 6, 2010....](#)
3. [Figure 1.3 Trade bar formation and properties.](#)
4. [Figure 1.4 Quote bar formation and properties.](#)
5. [Figure 1.5 Order messaging between the algorithm and brokerage.](#)
6. [Figure 1.6 Spread crossing: sell market orders fill at the bid price, and buy market orde...](#)
7. [Figure 1.7 Slippage costs are different between liquid and illiquid assets.](#)
8. [Figure 1.8 Security identification for a complex ticker change history.](#)
9. [Figure 1.9 Security identification for a simple ticker change history.](#)
10. [Figure 1.10 Stock value and number of shares owned before and after a split event.](#)

2. Chapter 2

1. [Figure 2.1 Running backtests in QuantConnect Cloud with debugging mode.](#)
2. [Figure 2.2 A data event occurs at the end of the data consolidation period. Tick data doe...](#)
3. [Figure 2.3 Overfitting and underfitting a model leads to poor predictions on samples outs...](#)

3. Chapter 4

1. [Figure 4.1 Sweetviz dashboard with no comparison target.](#)

2. [Figure 4.2 Sweetviz dashboard with gender as the comparison target.](#)
 3. [Figure 4.3 Box plot to identify outliers.](#)
 4. [Figure 4.4 Example of a non-stationary time series.](#)
 5. [Figure 4.5 Example of a stationary time series that results from differentiating a non-st...](#)
 6. [Figure 4.6 The original non-stationary time series before performing fractional different...](#)
 7. [Figure 4.7 The stationary time series that results from performing fractional differentia...](#)
 8. [Figure 4.8 Two non-stationary assets that are cointegrated have a stationary spread.](#)
 9. [Figure 4.9 Two cointegrated assets and their stationary spread.](#)
 10. [Figure 4.10 Correlation matrix of the target and factors.](#)
 11. [Figure 4.11 Feature importance plot of features 1 and 3. Feature 3 could be discarded to s...](#)
 12. [Figure 4.12 Importance plot of features 1 and 3 that were selected by recursive feature el...](#)
 13. [Figure 4.13 PCA transformed the feature matrix from three dimensions to two new dimensions...](#)
4. Chapter 5
1. [Figure 5.1 Example of linear regression.](#)
 2. [Figure 5.2 Example of polynomial regression.](#)
 3. [Figure 5.3 Example of LASSO regression.](#)
 4. [Figure 5.4 Example of LASSO regression coefficients.](#)
 5. [Figure 5.5 Example of ridge regression.](#)
 6. [Figure 5.6 Example of ridge regression coefficients.](#)
 7. [Figure 5.7 Example of Markov switching dynamic regression.](#)
 8. [Figure 5.8 Smoothed regime probabilities of the Markov switching dynamic regression.](#)
 9. [Figure 5.9 Example of decision tree regression.](#)
 10. [Figure 5.10 Decision tree structure.](#)
 11. [Figure 5.11 Example of SVM regression.](#)
 12. [Figure 5.12 Example of class distribution in a dataset.](#)
 13. [Figure 5.13 Example of confusion matrix.](#)

14. [Figure 5.14 Example of feature importance.](#)
 15. [Figure 5.15 Example of ROC curve.](#)
 16. [Figure 5.16 Example of hidden Markov model.](#)
 17. [Figure 5.17 ROC curve of Gaussian Naive Bayes.](#)
 18. [Figure 5.18 Example of neural network regression.](#)
 19. [Figure 5.19 Example of model loss.](#)
 20. [Figure 5.20 Example of true versus predicted ranking for multiple groups.](#)
 21. [Figure 5.21 Example of OPTICS clustering.](#)
 22. [Figure 5.22 Example of Amazon Chronos prediction.](#)
5. Chapter 6
1. [Figure 6.1 Example of trend identification using linear regression models.](#)
 2. [Figure 6.2 Equity curves of the strategy and selected benchmark.](#)
 3. [Figure 6.3 Predictions and observed log price.](#)
 4. [Figure 6.4 Scatter plot that colors each closing price based on the direction of future r...](#)
 5. [Figure 6.5 A bar chart showing the count of each label.](#)
 6. [Figure 6.6 Time series plots that show the probability of each label from the model...](#)
 7. [Figure 6.7 Time series plots that show the probability of each label from the model...](#)
 8. [Figure 6.8 Time series plots that show the probability of each label from the model...](#)
 9. [Figure 6.9 Time series plots that show the probability of each label from the model...](#)
 10. [Figure 6.10 The price and volatility of SPY, along with highlighted zones of low/high vola...](#)
 11. [Figure 6.11 Payoffs of a call option, put option, and an option straddle](#)
 12. [Figure 6.12 Equity curve, performance plots, and custom plots for Example 4.1.](#)
 13. [Figure 6.13 Monthly returns, crisis events, and sensitivity tests for Example 4.1.](#)

14. [Figure 6.14 Equity curve, performance plots, and custom plots for Example 4.2.](#)
15. [Figure 6.15 Monthly returns and crisis events for Example 4.2.](#)
16. [Figure 6.16 Equity curve, performance plots, and custom plots for Example 4.3.](#)
17. [Figure 6.17 Monthly returns and crisis events for Example 4.3.](#)
18. [Figure 6.18 Equity curve, performance plots, and custom plots for Example 5.](#)
19. [Figure 6.19 Monthly returns, crisis events, and sensitivity tests for Example 5.](#)
20. [Figure 6.20 Equity curve, performance plots, and custom plots for Example 6.](#)
21. [Figure 6.21 Monthly returns, crisis events, and sensitivity tests for Example 6.](#)
22. [Figure 6.22 Equity curve, performance plots, and custom plots for Example 7.](#)
23. [Figure 6.23 Monthly returns, crisis events, and sensitivity tests for Example 7.](#)
24. [Figure 6.24 Equity curve and performance plots for Example 8.1.](#)
25. [Figure 6.25 Monthly returns, crisis events, and sensitivity tests for Example 8.1.](#)
26. [Figure 6.26 Equity curve, performance plots, and custom plots for Example 8.2.](#)
27. [Figure 6.27 Monthly returns, crisis events, and sensitivity tests for Example 8.2.](#)
28. [Figure 6.28 Equity curve, performance plots, and custom plots for Example 8.3.](#)
29. [Figure 6.29 Monthly returns and crisis events for Example 8.3.](#)
30. [Figure 6.30 Asset representations in 3D space.](#)
31. [Figure 6.32 Asset representations in 3D space, color coded by group clusters.](#)
32. [Figure 6.34 Prices of candidate pairs and their normalized spread.](#)

33. [Figure 6.35 Equity curve, performance plots, and custom plots for Example 10.](#)
34. [Figure 6.36 Monthly returns, crisis events, and sensitivity tests for Example 10.](#)
35. [Figure 6.37 Equity curve, performance plots, and custom plots for Example 11.](#)
36. [Figure 6.38 Monthly returns, crisis events, and sensitivity tests for Example 11.](#)
37. [Figure 6.39 Plots of trading costs. Trading costs begin to decrease after the model has g...](#)
38. [Figure 6.40 Distribution of costs saved for delayed orders that filled before the time lim...](#)
39. [Figure 6.41 Distribution that shows the number of orders that filled at each delay time.](#)
40. [Figure 6.42 Distribution of costs saved for delayed orders that hit the time limit.](#)
41. [Figure 6.43 Equity curve, performance plots, and custom plots for Example 13.](#)
42. [Figure 6.44 Monthly returns, crisis events, and sensitivity tests for Example 13.](#)
43. [Figure 6.45 Architecture of the temporal CNN model.](#)
44. [Figure 6.46 Equity curve, performance plots, and custom plots for Example 14.](#)
45. [Figure 6.47 Monthly returns, crisis events, and sensitivity tests for Example 14.](#)
46. [Figure 6.48 Equity curve, performance plots, and custom plots for Example 15.](#)
47. [Figure 6.49 Monthly returns, crisis events, and sensitivity tests for Example 15.](#)
48. [Figure 6.50 Equity curve, performance plots, and custom plots for Example 16.](#)
49. [Figure 6.51 Monthly returns for Example 16.](#)
50. [Figure 6.52 Head and shoulder pattern and the typical entry timing.](#)

51. [Figure 6.53 Equity curve, performance plots, and custom plots for Example 17.](#)
52. [Figure 6.54 Monthly returns, crisis events, and sensitivity tests for Example 17.](#)
53. [Figure 6.55 Line plot of our synthetic H&S pattern.](#)
54. [Figure 6.56 Equity curve and performance plots for Example 18.1.](#)
55. [Figure 6.57 Monthly returns and crisis event for Example 18.1.](#)
56. [Figure 6.58 Equity curve and performance plots for Example 18.2.](#)
57. [Figure 6.59 Monthly returns and crisis event for Example 18.2.](#)
58. [Figure 6.60 Equity curve, performance plots, and custom plots for Example 19.1.](#)
59. [Figure 6.61 Monthly returns and crisis events for Example 19.1.](#)
60. [Figure 6.62 Equity curve, performance plots, and custom plots for Example 19.2.](#)
61. [Figure 6.63 Monthly returns and crisis events for Example 19.2.](#)
6. Chapter 7
 1. [Figure 7.1 The volatility “smile” and “skew”: implied volatil...](#)
 2. [Figure 7.2 Illustrative diagram of the continuous two-stage RL. The RL model can b...](#)
 3. [Figure 7.3 Daily closing price of AAPL \(Apple, Inc.\) shares and APPL Call options with 18...](#)
 4. [Figure 7.4 Convergence of loss function of the base model, that is, delta-mimicking stage...](#)
 5. [Figure 7.5 Convergence of the PnL-based penalty function \(loss function\) during the refin...](#)
 6. [Figure 7.6 Comparison of the decision evolution of the full model compared to the evoluti...](#)
 7. [Figure 7.7 Cumulative changes of the total wealth of the hedging portfolios of static hed...](#)
 8. [Figure 7.8 Cumulative changes of the net wealth of the hedged portfolio \(i.e., the option...](#)
7. Chapter 8
 1. [Figure 8.1 All these names are synonymous.](#)

2. [Figure 8.3 Average Eurusd return by time of day \(New York time\).](#)
3. [Figure 8.4 Equity curve of Primary trading strategy in out-of-sample period.](#)
4. [Figure 8.5 Equity curve of Corrective AI model in out-of-sample period.](#)
5. [Figure 8.6 Cumulative returns of strategies based on conditional vs. unconditional parame...](#)
6. [Figure 8.7 Training a portfolio performance prediction machine.](#)
7. [Figure 8.8 Live prediction of portfolio performance \(inferences\).](#)
8. [Figure 8.9 Optimization step.](#)
9. [Figure 8.10 Comparative performances of various portfolio optimization methods on Tech Por...](#)
10. [Figure 8.11 Cash allocation vs. market regime of tech portfolio.](#)
11. [Figure 8.12 Time evolution of allocations of TAA. portfolio \(GLD is deep blue.\)](#)

8. Chapter 9

1. [Figure 9.1 Prompt engineering complexity.](#)
2. [Figure 9.2 Intersection of the prompt components.](#)
3. [Figure 9.3 Amazon AWS architecture for LLM.](#)
4. [Figure 9.4 Amazon S3 dashboard.](#)
5. [Figure 9.5 Amazon Kendra wizard.](#)
6. [Figure 9.6 Amazon SageMaker wizard.](#)
7. [Figure 9.7 Snapshot from SageMaker Canvas.](#)

List of Tables

1. Chapter 8

1. [Table 8.1 Example trade ticks](#)
2. [Table 8.2 An 1-day slice of features table as input to the machine-learning model.](#)
3. [Table 8.3 Out-of-sample performances of unconditional vs. conditional parameter optimiza...](#)
4. [Table 8.4 A 1-day slice of features table as input to the neural network](#)
5. [Table 8.5 MESH ETF](#)

6. [Table 8.6 Tech Portfolio](#)
 7. [Table 8.7 Crypto portfolio](#)
 8. [Table 8.8 WSG's FX strategies portfolio](#)
 9. [Table 8.9 In- and Out-of-sample performance of TAA portfolio](#)
2. Chapter 9
1. [Table 9.1 Popular LLM by use case.](#)
 2. [Table 9.2 On-demand pricing on Amazon Web Services \(AWS\) for popular models.](#)
 3. [Table 9.3 RAG application monthly cost.](#)